CHAPTER 2:

Data Types & Operators



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Variables and Literals

- All **variables** must first be declared before they can be used. This involves stating the variable's type and name
- Eg int number1;
- A literal is the source code representation of a fixed value
- **Literals** are represented directly in code without requiring computation
- Eg

```
int decVal = 26;     // The number 26, in decimal
int octVal = 032;    // The number 26, in octal
int hexVal = 0x1a;// The number 26, in hexadecimal
```

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Types of Variable

- **Fields Variable** (Variables in a class)
  - Non-Static Field - Non-static fields are also known as *instance variables* because their values are unique to each *instance* of a class
  - Static Field - there is exactly one copy of this variable in existence regardless how many objects created from it.
- **Local variable** (Variables in method or block)
- **Parameter variable** (Variable in method declaration)

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Types of Variable

```
Public class Test
{
    int number1;
    public Test ( )
    {    int number4;
    }
    public void number ( int number2)
   {     int number3;
   }
}
```

→ **Field variable**

→ **Local variable**

→ **Parameter variable**

→ **Local variable**

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Types of Variable: Instance Variables

```
import java.io.*;

public class Employee{

   // this instance variable is visible for any child
class.

   public String name;

   // salary  variable is visible in Employee class
only.

   private double salary;

   // The name variable is assigned in the constructor.

   public Employee (String empName){

      name = empName;

   }
```

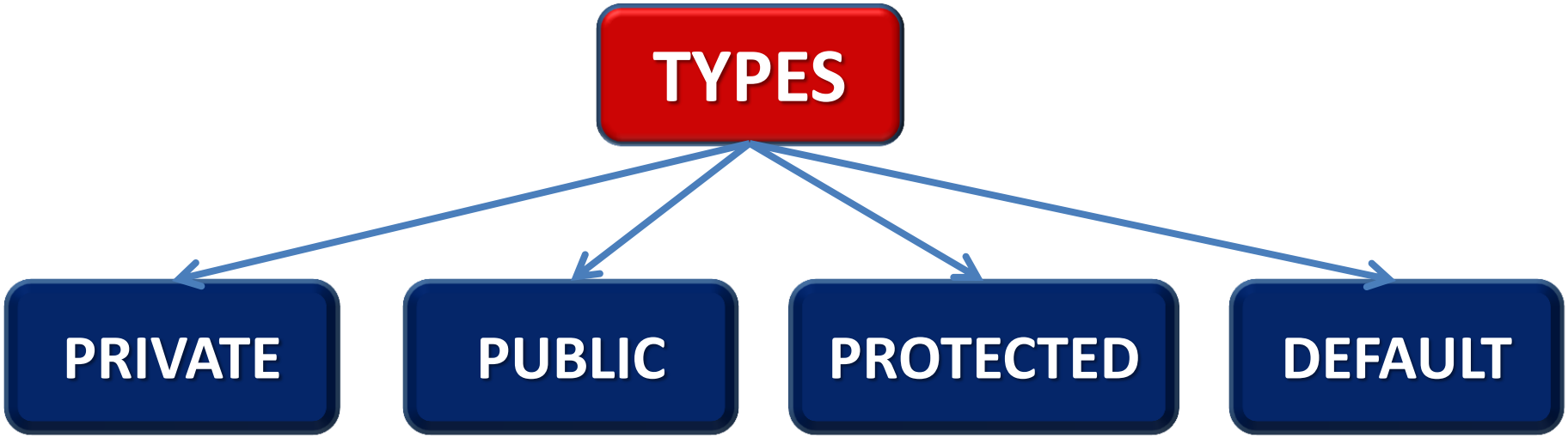
Types of Variable: Class Variables

```
import java.io.*;
public class Employee{
   // salary  variable is a private static variable
   private static double salary;
   // DEPARTMENT is a constant
   public static final String DEPARTMENT = "Development ";
   public static void main(String args[]){
      salary = 1000;
      System.out.println(DEPARTMENT + "average salary:" +
salary);
   }   }
```

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Access Modifier

- To set level of access for classes, fields or variables, methods and constructors.



Access Modifier

PRIVATE

- Most restrictive access level.
- Accessible within declared class.
- Class and interfaces cannot be private.

PUBLIC

Accessible from any other class in the java program.

Access Modifier

PROTECTED

- Variables, methods and constructors which are declared protected in a superclass can be accessed only by the subclasses in other package or any class within the package of the protected members' class.
- cannot be applied to class and interfaces.

DEFAULT

- Declare without access modifier.
- Available to any other class in the same package.

Variable Naming Rule

- Variable names are case-sensitive
- An unlimited-length sequence of Unicode letters and digits
- Always begin variable names with a letter, not "$" or "_“
- Subsequent characters may be letters, digits, dollar signs, or underscore characters

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Data Types (Primitive & Reference)

Data Type	Storing value	Default value
byte	8 bit signed integer	0
short	16 bit signed integer	0
int	32 bit signed integer	0
long	64 bit signed integer	0L
float	32 bit signed floating number	0.0f
double	64 bit signed floating number	0.0d
boolean	True or false	false
char	16-bit Unicode character	'\u0000'
String		null

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Format Specifier

Format	Data Type
%d	int, byte, short, long
%c	char
%f	Float, double
%s	String

System.out.format("Your username is %s and your password is %d", StrUsername, intnumber);

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Special Escape Sequence

	Description
\b	Backspace
\t	Tab
\n	Linefeed
\f	Formfeed
\r	Carriage return
\”	Display double quote
\’	Display Single quote
\\	Display back slash

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Strings

- Determine String length
- Concatenate String
- Converting String to numbers
- Converting number to string
- Change String value to uppercase and lowercase
- Trimming a String value
- Searching a character in a String
- Replacing character in a String
- Comparing String

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Examples

```
String original = "AbCD EfgH";
 String Strnumber = "123";
 String Str1, Str2, Str3, Str4;
 int Int1, Int2, Int3;

 Int1 = original.length();
 System.out.println(Int1); →Int1=9

 Str1 = "New" + original;
 System.out.println(Str1); →Str1 = NewAbCD EfgH

 Int2 = Integer.parseInt(Strnumber);
 System.out.println(Int2); →Int2 = 123

 Int3= 345;
 Str2 = Integer.toString(Int3);
 System.out.println(Str2); Str2= “345”
```

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Examples

```
String original = "AbCD EfgH";
 String Strnumber = "123";
 String Str1, Str2, Str3, Str4;
 int Int1, Int2, Int3;
```

```
Str3 = original.toUpperCase();
System.out.println(Str3); → Str3 = “ABCD EFGH”

Str3 = original.toLowerCase();
System.out.println(Str3); → Str3 = “abcd egh”

Str4 = original.trim(); //--> Leading and trailing white space remove

Str4 = Strnumber.replace("2", "2000");
System.out.println(Str4); →Str4 =  120003
```

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Operators

Operators	Examples
Assignment Operators	=
Arithmetic Operators	- ,+ ,*, /, %,
Unary Operator	+(Positive), -(Negative), ++ , --
Equality & Relational Operators	>, <, >=, <=, ==, !=
Conditional/Logical Operators	&&, ||
Bit wise Operator	>>, <<, &, ^, |
Compound Assignment Operators	+=, -=, *=, /=, %=, &=, ^=, >>=, <<=
Other comparison Operator	If varcat.equals("cat")

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Operators

Example Arithmetic Compound Assignment Operators:

int c = 3

c += 7 ⟶ **c = c + 7** ⟶ **c = 10**

c%= 17 ⟶ **c= c%17** ⟶ **c = (remainder)**

Increment and Decrement Operators

++a
a++
-- b
b - -

} Increase or decrease by 1

Reading Keyboard Input

- To read the value key in using keyboard
- Using the StreamReader class or Scanner class
- To use the Scanner class
  - Import java.util.Scanner
  - Initialize the object Scanner
  - Scan the keyboard input

Method 1

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Reading Keyboard Input

Method 1

```
import java.util.Scanner;

public class InputExp {

   public static void main(String[] args) {

       String name;
       int age;
       Scanner in = new Scanner(System.in);

       // Reads a single line from the console
       // and stores into name variable
       name = in.nextLine();

       // Reads a integer from the console
       // and stores into age variable
       age=in.nextInt();
       //in.close();

       // Prints name and age to the console
       System.out.println("Name :"+name);
       System.out.println("Age :"+age);

    }}
```

Reading Keyboard Input

```
import java.util.Scanner;
```

```
Scanner myscan = new Scanner(System.in);

String keyboardinput;

 //read string input using the object myscan and store the string in keyboardinput.

keyboardinput = myscan.nextLine();
```

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Reading Keyboard Input

Method	Description
public String next()	it returns the next token from the scanner.
public String nextLine()	it moves the scanner position to the next line and returns the value as a string.
public byte nextByte()	it scans the next token as a byte.
public short nextShort()	it scans the next token as a short value.
public int nextInt()	it scans the next token as an int value.
public long nextLong()	it scans the next token as a long value.
public float nextFloat()	it scans the next token as a float value.
public double nextDouble()	it scans the next token as a double value.

Dialog Boxes

Method 2

- import the Javax.Swing and Java.awt
- Use the JOptionPane

  JOptionPane.showInputDialog

  JOptionPane.showMessageDialog

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Dialog Boxes

```
import java.awt.*;

import javax.swing.*;

 String StrUsername;
```



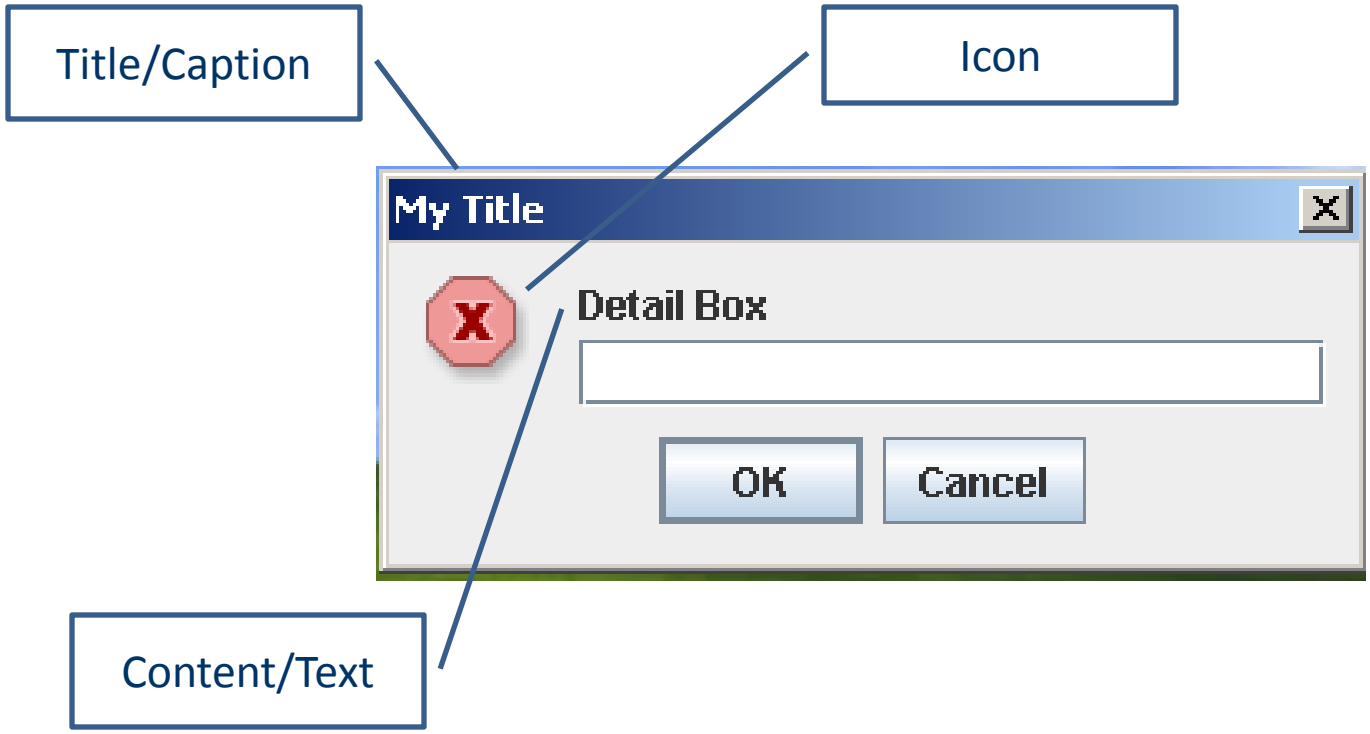

```
StrUsername = JOptionPane.showInputDialog(null,"Please key
in your Username“,"My Program",1);
```






```
 JOptionPane.showMessageDialog(null,"Your username is A”)
```

Dialog Boxes




```
JOptionPane.showInputDialog (null, "Detail Box", "My Title", JOptionPane.ERROR_MESSAGE);
```

Dialog Box Message Types

Constant Value	Icon
ERROR_MESSAGE	
INFORMATION_MESSAGE	
WARNING_MESSAGE	
QUESTION_MESSAGE	
PLAIN_MESSAGE	None

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Reading Keyboard Input using Dialog Box

```
import javax.swing.JOptionPane;
public class Addition {
   public static void main(String[] args) {
       String firstNumber;
       String secondNumber;
       int num1, num2, sum;

       firstNumber   =   JOptionPane.showInputDialog("Enter
                           first integer");
       secondNumber   =   JOptionPane.showInputDialog("Enter
                           second integer");
       num1 = Integer.parseInt(firstNumber);
       num2 = Integer.parseInt(secondNumber);
       sum = num1 + num2;

       JOptionPane.showMessageDialog(null,   "The   sum   is
  "+sum, "Results",      JOptionPane.PLAIN_MESSAGE);
       System.exit(0);
    }   }
```

Reading Keyboard Input using Dialog Box (2)

```
// read the string input and store in firstNumber.
firstNumber = JOptionPane.showInputDialog("Enter first integer");

// read the string input and store in secondNumber.
secondNumber = JOptionPane.showInputDialog("Enter second integer");

// convert the string in firstNumber to integer and store in num1
num1 = Integer.parseInt(firstNumber);

// convert the string in secondNumber to integer and store in num2
num2 = Integer.parseInt(secondNumber);
```

Formatting floating points

Example:

```
String Output;
Output = String.format("Hi %s. Your BMI is
%.2f", first,i/(j*j));
System.out.println(Output);
//OR
JOptionPane.showMessageDialog(null, Output,
"Results",JOptionPane.PLAIN_MESSAGE);
```

Reading character

```
char operator = input.charAt(0);
```